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2.a)

$$A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix} = \begin{bmatrix} 1+1 & 7 & 2 \\ 1 & 1+2 & 7 \\ 7 & 1 & 1+3 \end{bmatrix} = \begin{bmatrix} 2 & 7 & 2 \\ 1 & 3 & 7 \\ 7 & 1 & 4 \end{bmatrix}$$

$$\det(A) = 2(3.4 - 7.1) - 7(4.1 - 7.7) + 2(1.1 - 7.3)$$
$$= 2(12-7) - 7(4-49) + 2(1-21)$$
$$= 2(5) - 7(-45) + 2(-20)$$
$$= 10 + 315 - 40 = 285$$

b) $M_{11} = 3.4 - 7.1 = 5$

$M_{12} = 4.1 - 7.7 = -45$

$M_{13} = 1.1 - 7.3 = -20$

$M_{21} = 7.4 - 2.1 = 26$

$M_{22} = 2.4 - 7.2 = -6$

$M_{23} = 2.1 - 7.2 = -43$

$M_{31} = 7.7 - 3.2 = 43$

$M_{32} = 7.2 - 2.1 = 12$

$M_{33} = 3.2 - 7.1 = -1$

c.

$$C = \begin{bmatrix} +5 & +45 & -20 \\ -26 & -6 & +42 \\ +43 & -12 & -1 \end{bmatrix} \qquad C^T = \begin{bmatrix} 5 & -26 & 43 \\ +45 & -6 & -12 \\ -20 & 42 & -1 \end{bmatrix}$$

$$Adj(A) = C^T = \begin{bmatrix} 5 & -26 & 43 \\ 45 & -6 & -12 \\ -20 & 42 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{285} \cdot \begin{bmatrix} 5 & -26 & 43 \\ -45 & -6 & -12 \\ -20 & 42 & -1 \end{bmatrix} = \begin{bmatrix} 5/285 & -26/285 & 43/285 \\ -45/285 & -6/285 & -12/285 \\ -20/285 & 42/285 & -1/285 \end{bmatrix}$$

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e)

$$P = \begin{bmatrix} 120 & (50+17) & 200 \\ 80 & 1+2 & (10+17) \\ (30+17) & 90 & 170 \end{bmatrix} = \begin{bmatrix} 120 & 67 & 200 \\ 80 & 3 & 27 \\ 47 & 90 & 170 \end{bmatrix}$$

$$E = A \cdot P = \begin{bmatrix} 2 & 7 & 2 \\ 1 & 3 & 7 \\ 7 & 1 & 4 \end{bmatrix} \cdot \begin{bmatrix} 120 & 67 & 200 \\ 80 & 3 & 27 \\ 47 & 90 & 170 \end{bmatrix}$$

$E_{11} = 2.120 + 7.80 + 2.47 = 240 + 560 + 94 = 894$

$E_{12} = 2.67 + 7.3 + 2.90 = 134 + 21 + 180 = 335$

$E_{13} = 2.100 + 7.27 + 2.170 = 400 + 189 + 340 = 929$

$E_{21} = 1.120 + 3.80 + 7.47 = 120 + 240 + 329 = 689$

$E_{22} = 1.67 + 3.3 + 7.90 = 67 + 9 + 630 = 706$

$E_{23} = 1.200 + 3.27 + 7.170 = 200 + 81 + 1190 = 1471$

$E_{31} = 7.120 + 1.80 + 4.47 = 840 + 80 + 188 = 1108$

$E_{32} = 7.67 + 1.3 + 4.90 = 469 + 3 + 360 = 832$

$E_{33} = 7.200 + 1.27 + 4.170 = 1400 + 27 + 680 = 2107$

$$E = \begin{bmatrix} 894 & 335 & 929 \\ 689 & 706 & 1471 \\ 1108 & 832 & 2107 \end{bmatrix}$$

$P' = A^{-1} \cdot E$

$$= \frac{1}{285} \begin{bmatrix} 5 & -26 & 43 \\ -46 & -6 & -12 \\ -20 & 47 & -1 \end{bmatrix} \begin{bmatrix} 894 & 335 & 929 \\ 689 & 706 & 1471 \\ 1108 & 832 & 2107 \end{bmatrix}$$

$b = \frac{1}{285}$

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$P'_{11} = b(5.894 + (-26).689 + 43.1108) = b(4470 - 17914 + 47644)$

$= b.34200 = \frac{34200}{285} = 120$

$P'_{12} = b((5.335) + ((-26).706) + (43.832)) = b(1675 - 18356 + 35776)$

$= b.19095 = \frac{19095}{285} = 67$

$P'_{13} = b(5.929 + (-26).1471 + 43.2107) = b.(4645 - 38426 + 90601)$

$= b.57000 = \frac{57000}{285} = 200$

$P'_{21} = b(45.894 + (-6).689 + (-12)1108) = b(40230 - 4134 - 13296)$

$= b.22800 = \frac{22800}{285} = 80$

$P'_{22} = b(45.335 + (-6).706 + (-12).832) = b(15075 - 4236 - 9984)$

$= b.855 = \frac{855}{285} = 3$

$P'_{23} = b(45.929 + (-6)1471 + (-12)2107) = b(41805 - 8826 - 25284)$

$= b.7695 = \frac{7695}{285} = 27$

$P'_{31} = b((-20).894 + 47.689 + (-1)1108) = b(-17880 + 32383 - 1108)$

$= b.13395 = \frac{13395}{285} = 47$

$P'_{32} = b((-20).335 + 47.706 + (-1).832) = b(-6700 + 33182 - 832)$

$= b.25650 = \frac{25650}{285} = 90$

$P'_{33} = b((-20).929 + 47.1471 + (-1).2107) = b(-18580 + 69137 - 2107)$

$= b.48450 = \frac{48450}{285} = 170$

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Jadi $P' = \begin{bmatrix} 120 & 67 & 200 \\ 86 & 3 & 22 \\ 47 & 90 & 170 \end{bmatrix}$

Jadi $P' = P$ benar

Jika $\det(A) = 0$, maka A tidak memiliki inverS dan tidak ada nilai untuk dekripsi E